	Watts/cm^2	Decibels SPL	Example sound
	10^{-2}	140 dB	Pain
	10^{-3}	130 dB	
↑ Louder	10^{-4}	120 dB	Discomfort
	10^{-5}	110 dB	Jack hammers and rock concerts
	10^{-6}	100 dB	
	10^{-7}	90 dB	OSHA limit for industrial noise
	10^{-8}	80 dB	
	10^{-9}	70 dB	
	10^{-10}	60 dB	Normal conversation
	10^{-11}	50 dB	
	10^{-12}	40 dB	Weakest audible at 100 hertz
	10^{-13}	30 dB	
↓ Softer	10^{-14}	20 dB	Weakest audible at 10kHz
	10^{-15}	10 dB	
	10^{-16}	0 dB	Weakest audible at 3 kHz
	10^{-17}	-10 dB	
	10^{-18}	-20 dB	

TABLE 22-1
Units of sound intensity. Sound intensity is expressed as power per unit area (such as watts/cm^2), or more commonly on a logarithmic scale called *decibels SPL*. As this table shows, human hearing is the most sensitive between 1 kHz and 4 kHz.

The primary advantage of having *two* ears is the ability to identify the *direction* of the sound. Human listeners can detect the difference between two sound sources that are placed as little as three degrees apart, about the width of a person at 10 meters. This directional information is obtained in two separate ways. First, frequencies above about 1 kHz are strongly *shadowed* by the head. In other words, the ear nearest the sound receives a stronger signal than the ear on the opposite side of the head. The second clue to directionality is that the ear on the far side of the head hears the sound slightly *later* than the near ear, due to its greater distance from the source. Based on a typical head size (about 22 cm) and the speed of sound (about 340 meters per second), an angular discrimination of three degrees requires a timing precision of about 30 microseconds. Since this timing requires the volley principle, this clue to directionality is predominately used for sounds less than about 1 kHz.

Both these sources of directional information are greatly aided by the ability to turn the head and observe the change in the signals. An interesting sensation occurs when a listener is presented with exactly the same sounds to both ears, such as listening to monaural sound through headphones. The brain concludes that the sound is coming from the center of the listener's head!

While human hearing can determine the *direction* a sound is from, it does poorly in identifying the *distance* to the sound source. This is because there are few clues available in a sound wave that can provide this information. Human hearing weakly perceives that high frequency sounds are nearby, while low frequency sounds are distant. This is because sound waves dissipate their higher frequencies as they propagate long distances. Echo content is another weak clue to distance, providing a perception of the room size. For example,